Robert Ren (Ren He)

robert.renhe.work@gmail.com | github.com/renhe0707

EDUCATION

Purdue University West Lafayette, IN

Bachelor of Science in Applied Statistics | Minor: Management *Expected May 2027*

Relevant Coursework: Probability, Statistical Theory, Financial Mathematics, Linear Algebra, Time Series Analysis, Regression Analysis, Accounting

EXPERIENCE

Morgan Stanley Hong Kong SAR, China

Research Analyst Intern *Jun 2024 – Aug 2024*

- Analyzed 10+ multi-factor equity funds, computing Sharpe, Sortino, Max Drawdown, and Calmar ratios to build a performance-based ranking system directly applied to FOF portfolio quarterly rebalancing decisions
- Built Treynor-Mazuy factor attribution model (250-day rolling window) in Python to decompose stock selection alpha and market timing beta, identifying fund managers with persistent stock-picking ability
- Engineered automated data pipeline for financial data extraction, daily return computation, and factor exposure calculation (Python + pandas), improving quantitative report turnaround efficiency by 40%
- Produced 5+ quantitative research reports (alpha assessment, multi-factor attribution, risk decomposition) providing the basis for systematic fund screening criteria

Oin Auto San Francisco, CA

Data Analyst Intern *Dec 2023 – Mar 2024*

- Built classification model on 8,000+ records achieving 92% accuracy; designed automated data pipeline and operational dashboard supporting 50,000+ daily records for real-time processing and management decision-making

PROJECTS

Sector Rotation Strategy Research & Backtest *MA 37300 Financial Mathematics*

- Constructed sector prosperity composite scoring system across 31 Shenwan Level-1 industries using fundamental screening (ROE, revenue growth, PE percentile)
- Developed monthly sector rotation strategy in Python based on the scoring framework; 3-year backtest yielded 12.3% annualized return, outperforming CSI 300 by ~4.5 percentage points with max drawdown contained within 15%

Portfolio Risk Management & VaR Analysis *STAT 41600 Probability*

- Computed 95% VaR and CVaR for a multi-asset portfolio using Historical Simulation and Monte Carlo methods, validating model accuracy against 3 years of historical data
- Conducted stress tests simulating extreme market scenarios (2020 COVID crash, 2022 Fed rate hike cycle) to evaluate portfolio maximum loss exposure under tail risk events

Consumer Sector Equity Deep Dive *Independent Research*

- Conducted in-depth research on a consumer sector company covering business model, competitive landscape, and financial quality (ROE, cash flow, balance sheet structure), developing a complete investment thesis framework
- Performed valuation using DCF and comparable company analysis, issuing Buy/Hold rating with target price informed by industry cycle positioning; completed 8-page bilingual research report

LEADERSHIP

Bondminton at Purdue

Co-Founder *2025*

- Founded interdisciplinary badminton community from zero, growing membership to 39 within one semester; organized weekly sessions and facilitated cross-departmental networking

Purdue Student Government — Sustainability Committee

Director *Jan 2025 – Dec 2025*

- Authored 2 data-driven energy conservation proposals adopted and implemented by university facilities management

SKILLS

Programming: Python (pandas, NumPy, statsmodels, scikit-learn, matplotlib), R, SQL

Research Methods: Factor Models, Industry Research, Financial Analysis, Portfolio Optimization, Backtesting, Monte Carlo Simulation, Risk Measurement, Valuation Modeling (DCF / Relative Valuation)